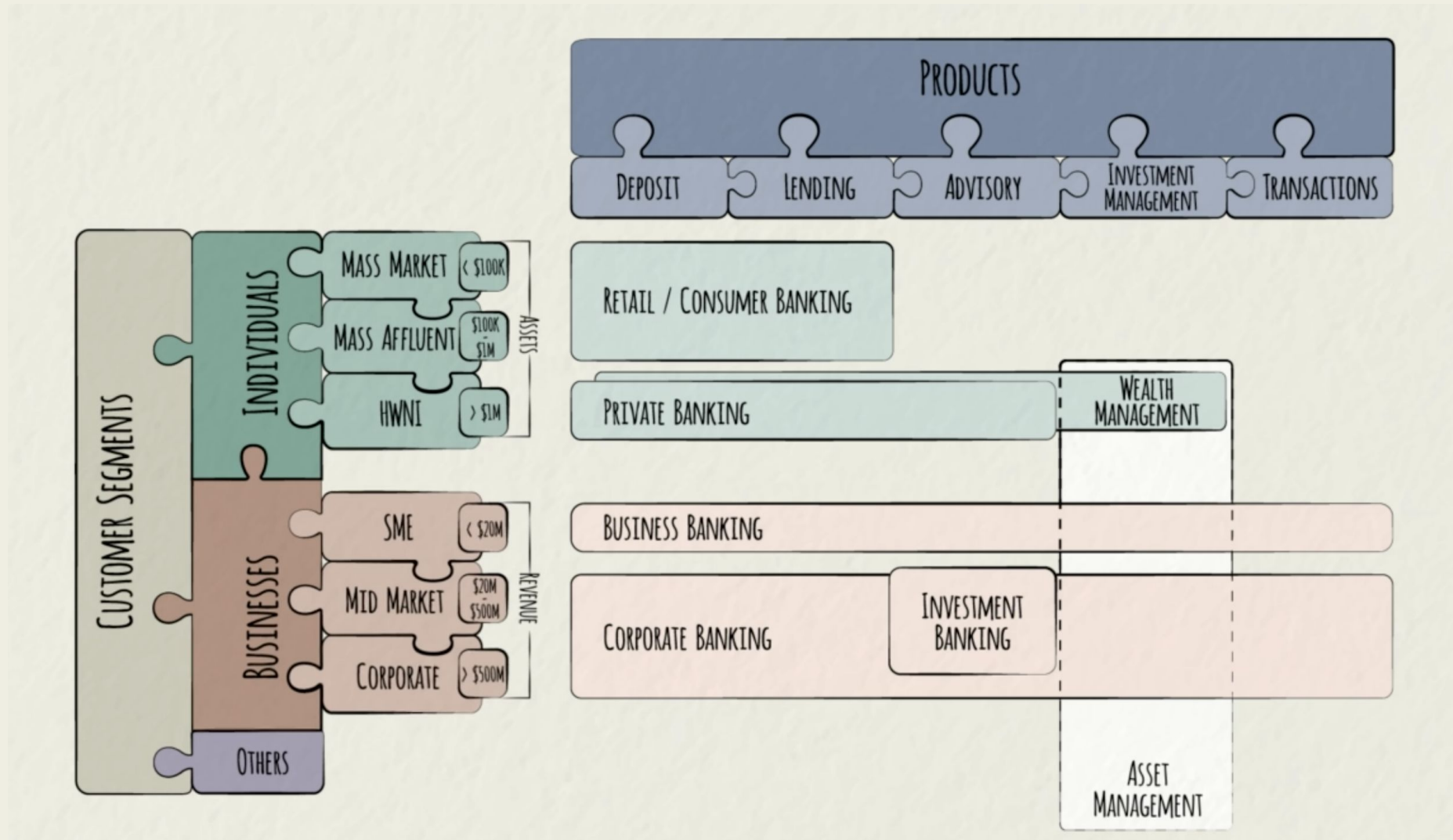


BANKING PRIMER



BUSINESS UNITS



BUSINESS UNITS

Customer Segments

Consumer	Business
Mass market – assets < \$100K	Small and medium enterprise (SME) – revenue <\$20M
Mass affluent – assets \$100K – \$1M	Mid market – revenue \$20M - \$500M
High net worth – assets > \$1M	Corporate – revenue >\$500M

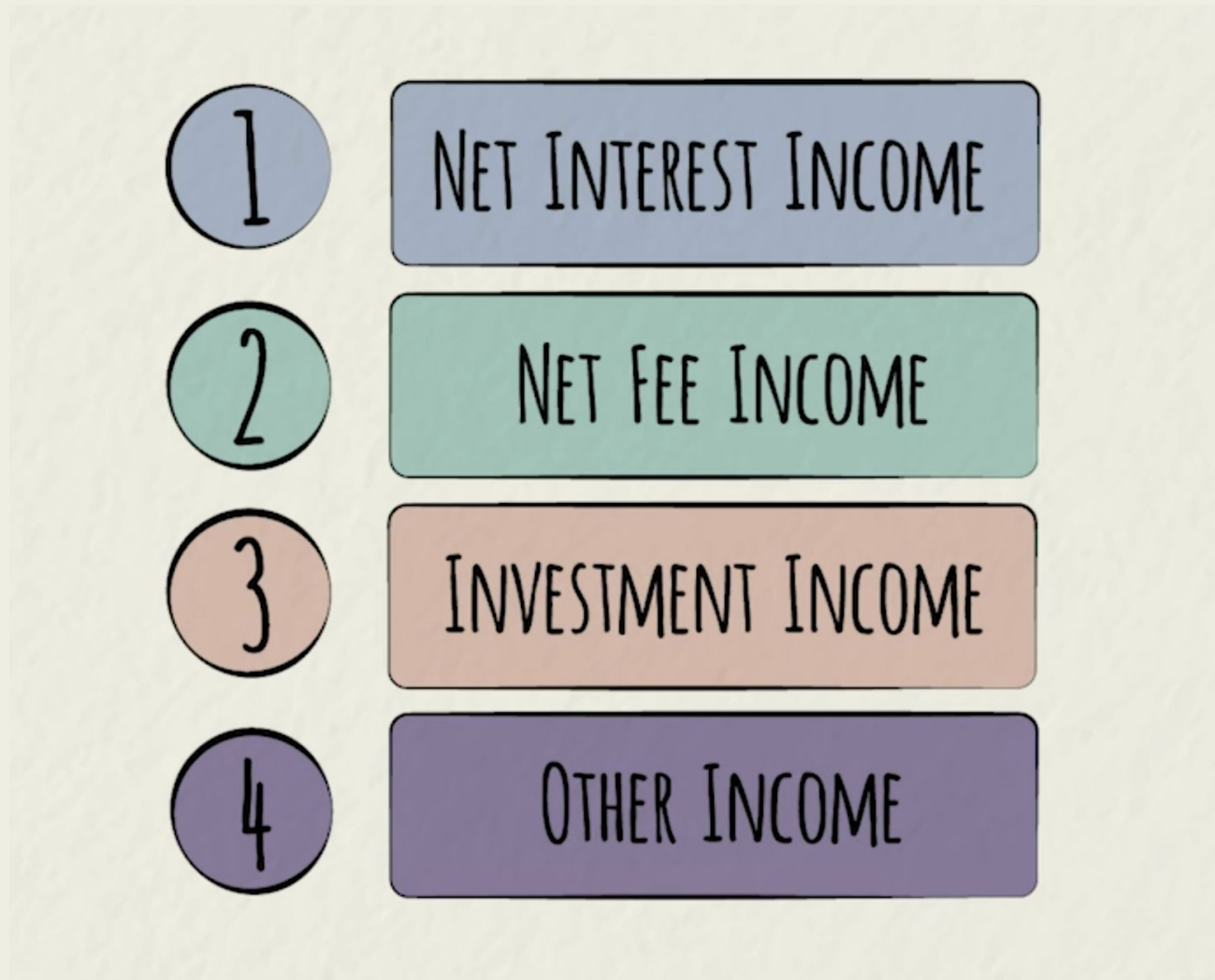
Banks may appear like a single entity, but are, in fact, composed of several business units:

- **Retail/consumer banking** serves individuals in the mass market and mass affluent segments, and offers deposit products such as checking and saving accounts, as well as lending products like credit cards, car loans, and mortgages
- **Private banking** and **wealth management** units serve high-net-worth individuals and offer advisory and investment management services in addition to deposit and lending products
- **Business banking** serves businesses in the small and medium-enterprise segment, offers deposit and lending products as well as transaction products such as point-of-sale machines to collect payments from customers and an automated payment system to pay vendors and employees
- **Corporate banking** serves businesses in the mid market and corporate segments, and offers additional advisory and investment management options to clients. Companies looking to go public can rely on a bank's **investment banking** unit to help them structure the terms of their bond or stock issuance and find potential investors

Not all banks have all of these units:

- Banks that take deposits and offer lending products to individuals and businesses, e.g. Capital One, are called **commercial banks**
- Banks that primarily offer investment banking services, e.g. Morgan Stanley and Goldman Sachs, have historically been referred to as **investment banks**
- Banks such as JP Morgan Chase, Bank of America, HSBC and Deutsche Bank, that offer both deposit/lending and investment banking services are referred to as **universal banks**

BUSINESS MODEL – INCOME SHEET



BUSINESS MODEL – INCOME SHEET

Banks generate four types of revenue:

- **Net interest income** is the difference between **interest income** and **interest expense** (the interest banks charge customers and the interest they pay to source the funds)
 - Often the largest total source of revenue (frequently between 50% and 70%)
 - Banks diversify and add other revenue streams because a) net interest income is capped by regulations on how much banks can loan and b) net interest income is sensitive to changes in interest rates
- **Net fee income** is income generated from fees charged to customers
 - e.g., overdraft fees, annual credit card fees, or transaction fees
 - Net fee income on the income statement refers to the sum of fees collected minus the associated fees paid to third parties
- **Investment income** is managed by the bank's treasury unit (markets division) for the bank as well as the bank's clients
 - Can come in the form of dividends or the change in the value of investments
 - Trading tends to account for a smaller share of income for commercial banks compared to global universal banks
 - Heavily regulated to prevent banks from taking excessive risk
- **Other income** can vary depending on the activities that the bank is involved in
 - e.g., commissions from insurance sales and rental income from properties owned

BANK BALANCE SHEET

SOURCE OF FUNDS

LIABILITIES

+

EQUITY

=

USE OF FUNDS

ASSETS

BANK BALANCE SHEET

The balance sheet provides a high-level picture of how a bank sources and uses its funds

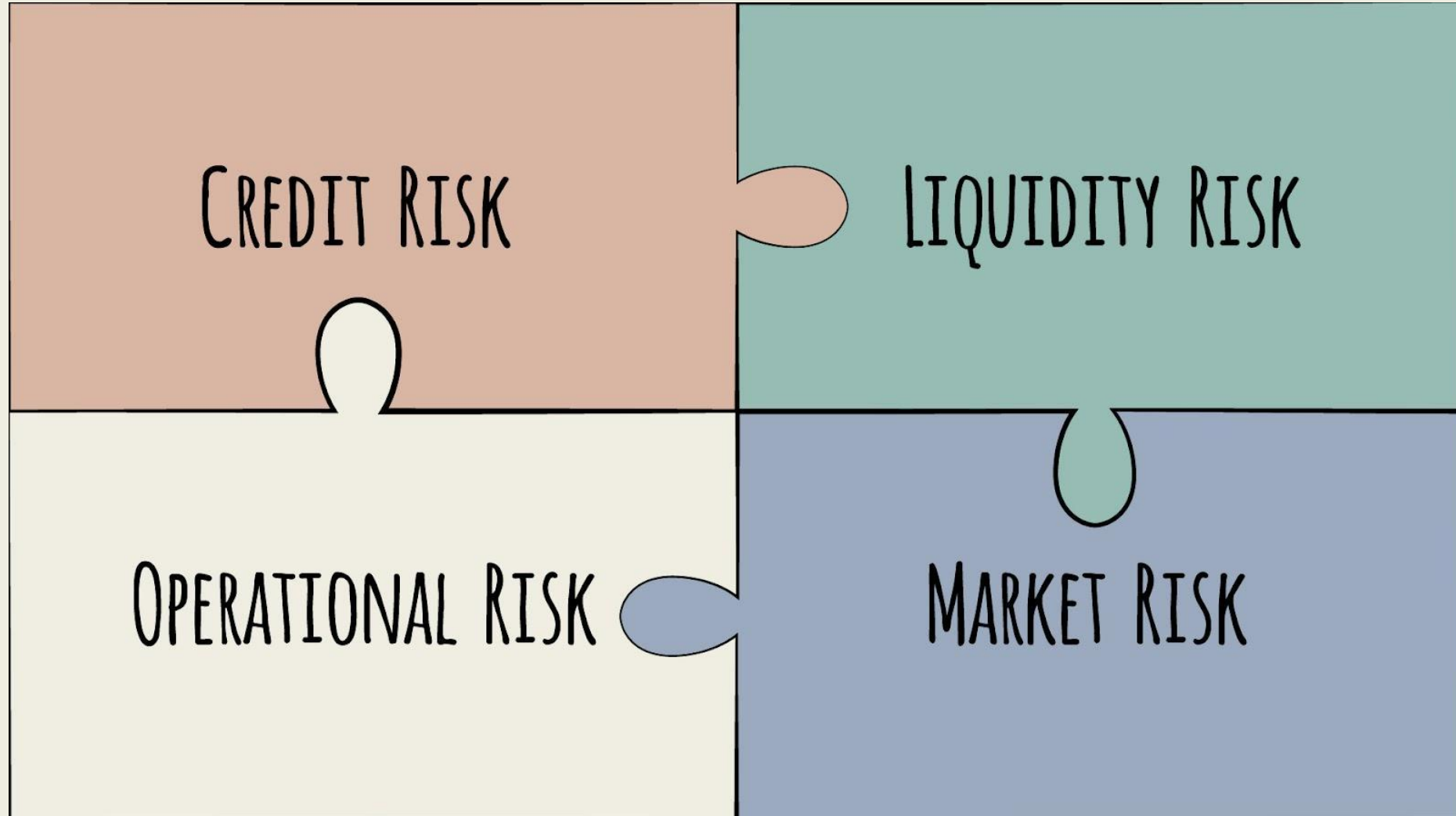
Source of funds:

- **Equity** is the portion of funds owned by the bank's shareholders
- **Liabilities** are the funds sourced from customers and investors through cash deposits and bonds

Use of funds:

- **Assets** represent the total funds that can be put to work and are always equal to a bank's liabilities and equity
- They can be deployed in four ways:
 1. Cash, which is low risk and low return and, therefore, not highly preferred. Banks are required to keep a portion of customer deposits, often 10%, available in cash at all times. This is called the **reserve requirement**
 2. Loans, which frequently represent the bulk of a bank's assets on the balance sheet
 3. Investments, which are often much smaller than loans since they are considered riskier, and regulators limit the amount of risk a bank can take
 4. Non-financial assets such as property and equipment

RISKS



RISKS

There are four types of risks that a bank has to manage:

- **Credit risk** is the risk that borrowers will default on their debt
 - A process called underwriting assesses customers' credit risk. Underwriting takes in key information such as income, total debt, and credit history and returns a yes/no decision, the loan amount, price, and mitigation measures like a collateral requirement
 - Collateral is an asset that the borrower pledges to the lender to secure the loan. The underwriting process determines what loans will end up in the bank's loan portfolio
 - If a bank chooses to lend to riskier borrowers to generate higher interest income, it would likely experience higher default rates and credit losses. Banks actively manage this risk-reward trade-off depending on economic cycle, customer behavior, and risk appetite
 - Each bank has a different underwriting process and it is a competitive differentiator for banks
- **Liquidity risk** is the risk that the bank won't be able to obtain enough cash when needed to meet its financial obligations
 - These obligations can arise both on the liability side, such as a bond repayment or a large customer withdrawal, or on the assets side, like a loan commitment or cash needed to meet the reserve requirement
 - Liquidity risk is monitored on a daily basis by the bank's risk team

RISKS

- **Market risk** is the risk that market conditions will go against the bank. Examples could include:
 - An increase in interest rates, raising the cost of funding for a bank. Banks put mitigation measures in place to make sure changes in the cost of funding do not have a major impact on earnings
 - A drastic change in security prices, like a stock market crash. An event like this would have three consequences:
 - First, the bank will incur losses on the securities it holds as investments
 - Second, since clients use securities as collateral for their loans, a decrease in the security value will increase the credit risk of bank's loan portfolio
 - Third, because banks use securities as collateral for their own borrowing, they would have to show additional assets as collateral
- **Operational risk** is the risk of any system or human error that can lead to losses. Examples include fraud or clerical errors by loan officers or even robbery

KEY RATIOS

There are several ratios used to monitor a bank's performance. They fall into three categories: financial performance, asset quality, and balance sheet strength

Financial performance – Shows shareholders how efficient and profitable the bank is

- The Cost-to-Income ratio (CI) is an indicator of operational efficiency that shows how much of the operating income goes to covering operational expenses like salaries and marketing. It's calculated by dividing operating expenses by operating income and typically ranges between 40 to 60 percent for commercial banks
- Net Interest Margin (NIM) is the interest margin the bank earns on its loans and is a measure of profitability and effective use of assets. It is calculated as net interest income divided by total loans. If a bank's NIM is lower than its peers, the bank might be undercharging or it has a higher cost of funds
- Return on Assets (ROA) is a measure of how effectively the bank uses its funds or in other words, assets, to generate income. It is calculated as net income divided by total assets
- Return on Risk-Weighted Assets (RORWA) is used to adjust ROA for the risk undertaken by the bank, by weighing assets based on their risk level

KEY RATIOS

Asset quality – A measure of how effectively a bank uses its funds or in other words, assets, to generate income

- The Loan Loss Provision is the expected cost of a borrower not repaying their loan. Funds set aside to cover this cost appear as an expense on the income statement
- A Nonperforming Loan (NPL) is one that is more than 90 days past due and regarded as unlikely to be repaid
- The NPL ratio shows the share of loans that are nonperforming. A high NPL ratio could be alarming because it is an indicator of low asset quality and, therefore, high credit risk
- A charge-off is how the bank categorizes a loan it is almost “certain” the borrower won’t repay. At this point, the bank will stop trying to collect and start the recovery process through liquidation of collateral or by selling the loan to other collection agencies
- Net charge offs (NCO) is the charge-off amount minus any recoveries
- The NCO rate is the total NCO amount relative to the bank’s total loan portfolio. It is an indicator of the bank's ability to recover on its loans. A high NCO rate relative to peers would be a concern for the bank’s outlook
- The Loan Loss Provisions Coverage Ratio (LLPCR or PCR) is a measure of how well the bank can withstand future losses by looking at its income relative to the charge-offs. It is calculated by dividing the sum of pretax income and loan loss provisions by net charge-offs

KEY RATIOS

Banks generally fail for one of two reasons:

- Insolvency – It owes more than it owns (i.e., liabilities > assets)
- Illiquidity – It isn't able to raise enough cash quickly to pay its immediate obligations. To avoid each scenario, regulators set two groups of indicators:

Capital Adequacy – A bank would be called insolvent if all capital is gone and its assets are worth less than its liabilities. Regulators require banks to hold enough capital to protect debt holders such as depositors from losses

- Capital adequacy ratio (CAR) tracks the capital relative to a bank's assets. CAR is calculated as total capital divided by risk-weighted assets to adjust for the risk level in the portfolio. While national approaches may vary, most countries follow the regulatory capital guidelines called Basel 3, which requires CAR to be at least 10.5%

Liquidity – Several ratios help banks' executives actively manage their ability to meet obligations as they come due

- The Loans-to-Deposits ratio (LDR) is the ratio of the total loans issued by the bank to total deposits it has collected. While a high LDR suggests an efficient deployment of deposits, it also means that the bank is exposing itself to liquidity risk if depositors suddenly withdraw. Most commercial banks manage their LDR in the 70-90% range
- Liquidity Coverage Ratio (LCR) is another indicator to ensure that the bank holds enough liquid assets to cover cash outflows over a potentially stressed period of 30 days. LCR is calculated by dividing total high-quality liquid assets by net outflows and should be at least 100%

BANKING SYSTEM

Credit

An economy grows as a result of increased economic activity, where consumer spending encourages businesses to invest more, which creates more employment, which drives consumer spending, and so on. In an expansionary period, banks will have higher confidence that borrowers can repay their debt, so they will lend more, which will provide more credit to the economy

- In the fractional reserve banking system, credit is expanded by allowing banks to keep only a fraction of deposits as cash and issuing the rest as loans. This is why credit, not money, is the fuel that every economy runs on, because it's primarily what drives investments and the spending necessary for growth

Central Bank

Central banks are one of the most influential institutions in any country and they play three very critical roles:

- Lender of last resort – If commercial lenders are unwilling to extend credit to the bank, the Central Bank steps in as the lender of last resort. Commercial banks try to avoid borrowing from the Central Bank, as doing so implies that the bank is having financial troubles

BANKING SYSTEM

- Money supply – Central banks are in charge of regulating the credit supply to the economy. While increasing credit will stimulate the economy, too much of it may lead to excessive borrowing, high inflation, and asset bubbles, so central banks employ a range of tools to maintain the credit supply at an ideal level
 - The most important of these is the interest rate. In the interbank lending market, the interest rate is set based on the demand and supply from banks
 - Central banks can move the benchmark interest rate through a monetary tool called open market operations. For example, if the central bank wants to expand credit, it injects liquidity into the system by buying government bonds. This will lower the demand for liquidity and therefore, the interest rates, which has two effects. First, lower funding costs will lead to cheaper loans, which mean higher demand for loans. It also reduces the return on cash, which motivates lenders to issue more loans
- Financial stability – Given the banking system's critical role, a systemic malfunction would be catastrophic for any economy. Using ratios and other indicators, central banks monitor commercial banks' financial health and the stability of the overall system. They can issue mandatory corrective orders to banks such as increasing capital or de-risking loan portfolios. They can even intervene and close down a bank if it's deemed a threat to depositor funds or the stability of the system

GLOSSARY

- Charge off – Category of loan that a bank is almost certain the borrower won't repay
- Collateral – An asset that the borrower pledges to the lender to secure the loan
- Commercial banks – Banks that take deposits and offer lending products to individuals and businesses
- Credit risk – The risk that borrowers will default on their debts
- Illiquid – The inability to raise enough cash quickly to pay immediate obligations
- Insolvent – Having greater liabilities than assets
- Investment banks – Banks that primarily offer investment banking services
- Liquidity risk – The risk that the bank won't be able to obtain enough cash when needed to meet its financial obligations
- Net charge-offs (NCO) – The difference between the charge-off amount and any recoveries the bank can make on the loan
- Net fee income – The income generated from fees charged to customers
- Net interest income – The difference between the interest that banks charge customers and the interest it pays to source funds
- Nonperforming Loan (NPL) – Designation given to a loan when payment is >90 days overdue
- Reserve requirement – The part of customer deposits, often 10%, which banks are required to keep available in cash to provide a buffer in case there is a sudden withdrawal from the customers
- Underwriting – A process used to assess credit risk, using a customer's key information such as income, debt, and credit history
- Universal banks – Banks that operate both commercial and investment units